



AWS Content Delivery and DNS Services

[Home](#) » [AWS Cheat Sheets](#) » AWS Content Delivery and DNS Services

This is a category of AWS

services include services for caching



ing content around the world and providing intelligent Domain Name System (DNS) services for your applications.

Amazon Route 53

Amazon Route 53 is the AWS Domain Name Service.

Route 53 performs three main functions:

- Domain registration – Route 53 allows you to register domain names.
- Domain Name Service (DNS) – Route 53 translates name to IP addresses using a global network of authoritative DNS servers.
- Health checking – Route 53 sends automated requests to your application to verify that it's reachable, available, and functional.

You can use any combination of these functions.

Route 53 benefits:

- Domain registration.
- DNS service.
- Traffic Flow (send users to the best endpoint).
- Health checking.
- DNS failover (automatically change domain endpoint if system fails).
- Integrates with ELB, S3, and CloudFront as endpoints.



Routing policies determine how Route 53 DNS responds to queries.

The following table highlights the key function of each type of routing policy:

Policy	What it Does
Simple	Simple DNS response providing the IP address associated with a name
Failover	If primary is down (based on health checks), routes to secondary destination

Geolocation	Uses geographic location you're in (e.g. Europe) to route you to the closest region
Geoproximity	Routes you to the closest region within a geographic area
Latency	Directs you based on the lowest latency route to resources
Multivalued answer	Returns several IP addresses and functions as a basic load balancer
Weighted	Uses the relative weights assigned to resources to determine which to route to

Amazon CloudFront

Amazon CloudFront is a content delivery network (CDN) that allows you to store (cache) your content at "edge locations" located around the world.

This allows customers to access content more quickly and provides security against DDoS attacks.

CloudFront can be used for data, videos, applications, and APIs.

CloudFront benefits:

- Cache content at Edge Location for fast distribution to customers.
- Built-in Distributed Denial of Service (DDoS) attack protection.
- Integrates with many AWS services (S3, EC2, ELB, Route 53, Lambda).

Origins and Distributions:

- An origin is the origin of the files that the CDN will distribute.
- Origins can be either an S3 bucket, an EC2 instance, an Elastic Load Balancer, or Route 53 – can also be external (non-AWS).
- To distribute content with CloudFront you need to create a distribution.

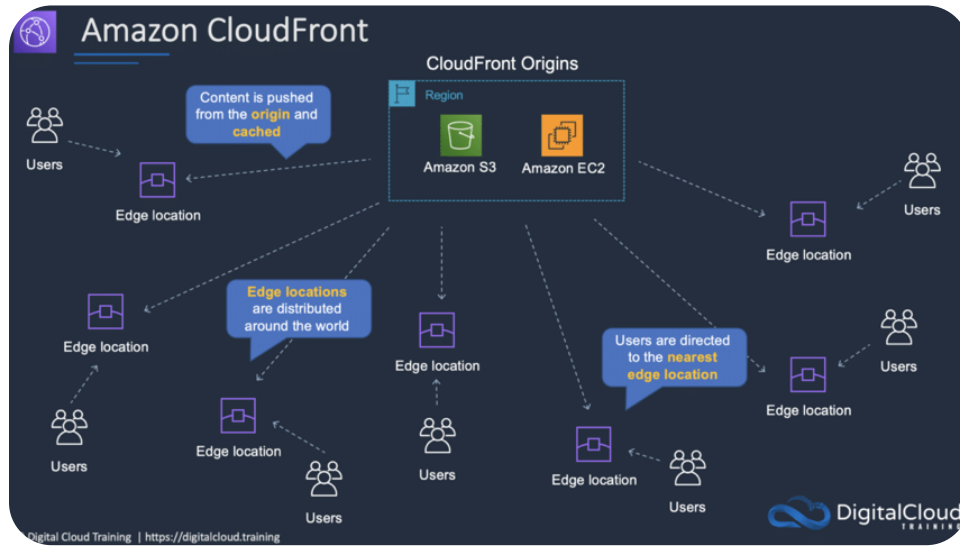


CloudFront uses Edge Locations and Regional Edge Caches:

- An edge location is the location where content is cached (separate to AWS regions/AZs).
- Requests are automatically routed to the nearest edge location.
- Regional Edge Caches are located between origin web servers and global edge locations and have a larger cache.
- Regional Edge caches aim to get content closer to users.

The diagram below shows where Regional Edge

Caches and Edge Locations are placed in relation to end users:



Related posts:



PREVIOUS
AWS Cloud Computing Concepts

NEXT
AWS Monitoring and Logging Ser...

AWS Training

AWS

Find Answers

Connect

Live Virtual Bootcamps	Certifications	Getting Started with AWS	About us
Monthly Yearly Plans	AWS Cloud Practitioner	Knowledge Hub	Newsletter
Hands-on Challenge Labs	AWS Solutions Architect	Cheat Sheets	Contact us
Training for Businesses	AWS Developer Associate	FAQ	Submit Feedback
AWS Books for Offline Study	AWS CloudOps Engineer	Join our Slack Channels	Join our Team
	AWS Solutions Architect PRO		

Get the Free Beginner's Guide to AWS Certification

First name*

Last name*

Email*

Uni 

+1

By submitting this form, you agree to receive SMS and emails from Digital Cloud Training, as outlined in our [Privacy Policy](#). You can unsubscribe at any time by replying STOP or using the unsubscribe link.

Access Free Guide

Terms

- Terms of Service
- Privacy Policy
- Refund Policy
- Sitemap

©2025 Digital Cloud Training